

Measuring how fast prices incorporate scheduled macro releases

The model

proj-price-discovery

Regenerated by just model. Do not edit the PDF.

This document states the model behind the estimand fixed in ADR 0002, and the reasoning that selected it. It is written for a reviewer deciding whether the measurement is sound. The corresponding source of truth is `docs/framings/`; where the two disagree, that directory wins.

Scope. This is a measurement study. There is no trading strategy here, no backtest and no signal. The quantity of interest is the speed at which a price converges after a scheduled information event.

1 The question and the evidence that it is answerable

US macro releases arrive at a known second and are not caused by the market. CPI and the Employment Situation are published at 08:30 ET. Prices come from Binance, which trades continuously, so a release lands in an open market with no auction and no session boundary to model.

The premise was tested before any modelling. Figure 1 shows BTCUSDT around 08:30 ET on nine CPI release dates against the prior weekday at the same clock time. In 9 of 9 dates the largest one-second move in the 30 seconds after the release exceeds the comparison day's, by a factor of 3.6 to 64, median 5.6.

2 Notation

Symbol	Meaning
$e \in \{1, \dots, E\}$	event index; one scheduled release, $E = 214$
τ	event time, seconds since the release instant
$R_e(\tau)$	cumulative log return from the last price strictly before release
H	terminal horizon, 1 hour
$R_e(H)$	the eventual move
$\phi_e(\tau)$	fraction of the eventual move realised by τ
A_e	amplitude, basis points
λ_e	incorporation rate, s^{-1}
μ, σ	population mean and standard deviation of $\log \lambda_e$

3 The estimand

$$\phi_e(\tau) = \frac{R_e(\tau)}{R_e(H)} \tag{1}$$

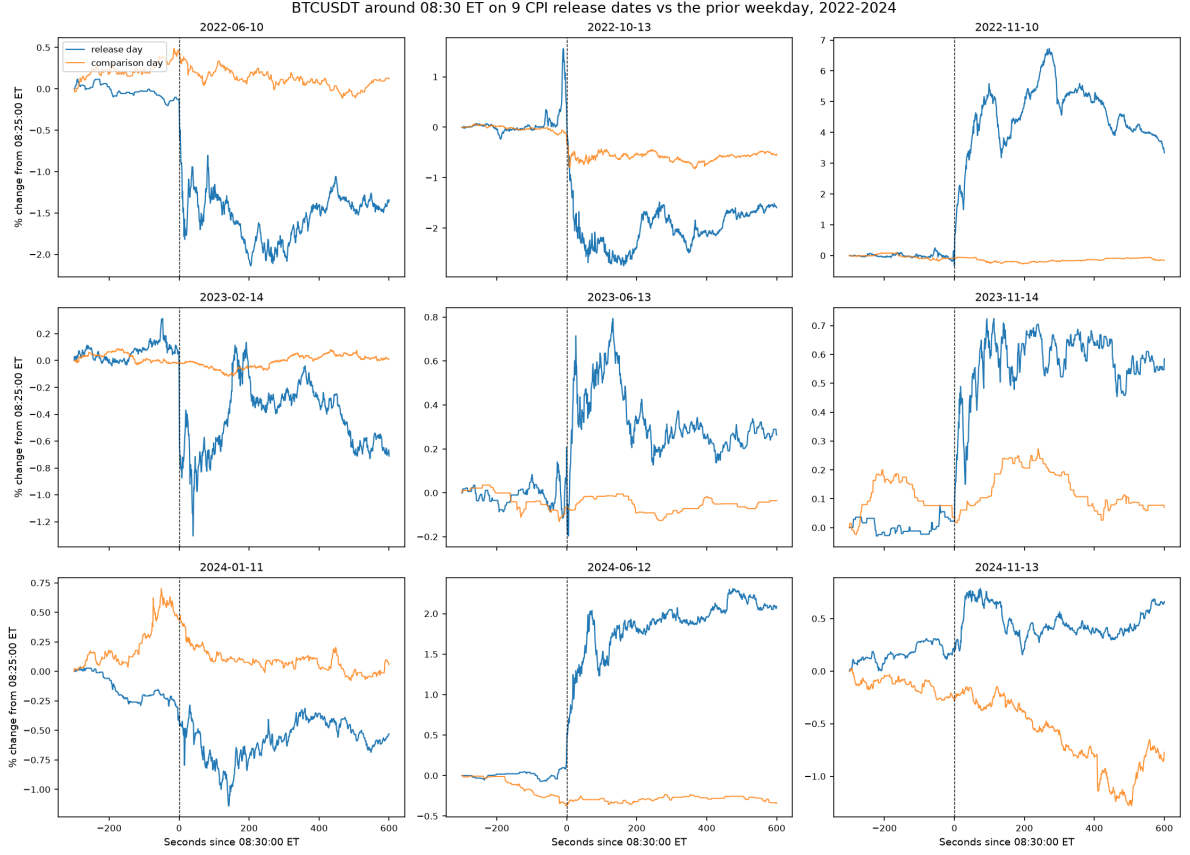


Figure 1: Kill-check. Release day against the prior weekday at the same clock time. Direction is inconsistent across releases, which is expected: the sign follows the surprise. Two panels (2022-11-10 and 2023-06-13) overshoot and revert within the window, which is the failure mode Section 9 treats as a testable claim rather than an assumption.

reported at $\tau \in \{1\text{ s}, 10\text{ s}, 1\text{ min}, 10\text{ min}, 1\text{ h}\}$ with credible intervals, and summarised by a half-life in seconds.

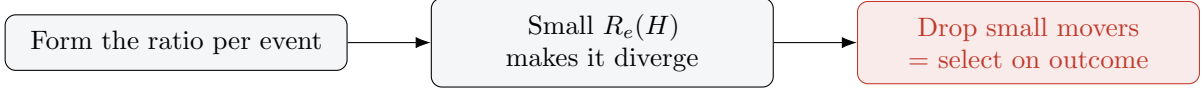
Note what is being claimed. $\phi_e(H) = 1$ by construction, because H defines the destination. The study measures the *speed* of convergence and makes no claim that the destination is the efficient price.

4 Why the obvious estimator is not used

The direct route is to compute $R_e(\tau)/R_e(H)$ per event and average. It has a closed form and costs nothing to evaluate. It is also unusable.

$R_e(H)$ appears in a denominator. For a release that moved the price little, $R_e(H)$ is a small number measured with error, and the ratio diverges. A single event whose eventual move was 2 bp against 3 bp of measurement noise can dominate the mean and flip its sign.

The standard repair is to drop events whose eventual move was small. That repair **conditions on the outcome**: it selects events using a quantity observed after the release, unavailable ex ante. It is the most likely route to a clean-looking wrong answer in this design, and it is what a reviewer should look for first.



5 The model

5.1 One event

Assume the price approaches its eventual level exponentially:

$$\mathbb{E}[R_e(\tau)] = A_e \left(1 - e^{-\lambda_e \tau}\right), \quad \text{so} \quad \phi_e(\tau) = 1 - e^{-\lambda_e \tau} \quad (2)$$

Two parameters with distinct roles. A_e is where the price is heading and varies across events because surprises vary in size; it is a nuisance parameter. λ_e is how fast it gets there and is the object of interest.

The separation is what removes the division. In (2) the amplitude cancels *algebraically*, so ϕ_e never touches A_e numerically. An event with a small amplitude contributes weak evidence about λ_e , which a diffuse posterior represents correctly, rather than a divergent ratio, which nothing represents correctly.

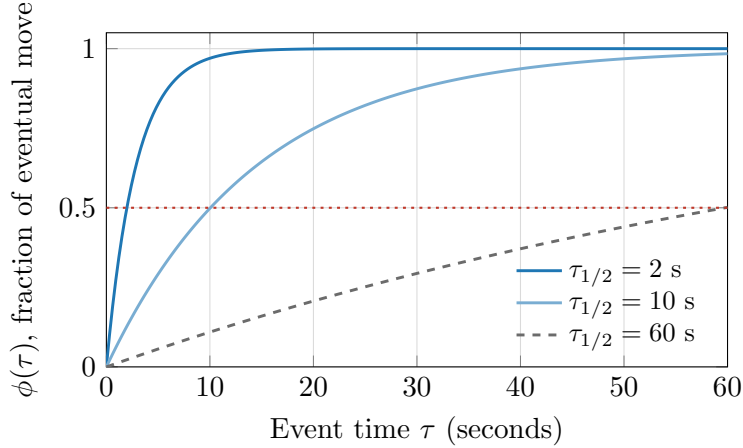


Figure 2: The incorporation curve at three rates. The reported summary is the half-life $\tau_{1/2} = \ln 2 / \lambda$, the seconds by which half the eventual move has occurred, because seconds are a unit a reader can check against intuition and s^{-1} is not.

5.2 Observation model

$$R_e(\tau_k) \sim \mathcal{N}\left(A_e \left(1 - e^{-\lambda_e \tau_k}\right), \zeta^2 \tau_k\right) \quad (3)$$

The variance grows with the horizon. Over one second little happens beyond the release; over one hour a great deal of unrelated news arrives. Homoscedastic errors would assert that an hour-long return is as precise a measurement as a one-second return, and hour-scale noise would then drag the estimate of λ_e .

Variance proportional to τ follows from independent increments in the background process. It is the default and it is an assumption.

5.3 Pooling across events

$$\log \lambda_e \sim \mathcal{N}(\mu, \sigma^2) \quad (4)$$

Each event keeps its own rate, and those rates are themselves drawn from a shared population. Table 1 states why neither extreme is viable at $E = 214$.

	What it assumes	Why it fails here
No pooling	Events share nothing	Per-event posteriors too diffuse to report
Complete pooling	Events are identical	Forecloses release-type and year contrasts
Partial pooling	Rates share a distribution	—

Table 1: Pooling choices. Partial pooling is a consequence of the sample size, not a preference.

Under (4) each λ_e is shrunk toward μ in proportion to how noisy that event’s data is. Approximately, writing v_e for the variance of the event’s own estimate,

$$\hat{\lambda}_e \approx \frac{\hat{\lambda}_e^{\text{own}}/v_e + \mu/\sigma^2}{1/v_e + 1/\sigma^2} \quad (5)$$

so a precisely measured event keeps its own estimate and a noisy one defaults toward the population. The weighting is not tuned; it follows from the two variances.

σ is inferred, not fixed. If events genuinely differ, the data forces σ large, shrinkage weakens, and the model approaches no pooling. If they do not, σ contracts and it approaches complete pooling. The degree of pooling is determined by the data.

The logarithm in (4) is not cosmetic. Rates are positive and plausibly span orders of magnitude. A normal prior on λ_e directly would place mass on negative rates and would treat $0.1 \rightarrow 0.2$ as a step of the same size as $10.1 \rightarrow 10.2$, when the first is a doubling and the second is a rounding error.

Amplitudes take the same treatment, $\log |A_e| \sim \mathcal{N}(\mu_A, \sigma_A^2)$ with the sign free, since a surprise may push the price either way.

6 The model as a factor graph

Figure 3 draws the joint distribution. Circles are random variables, filled circles are observed, small squares are factors, and the plate repeats its contents over the index in its corner.

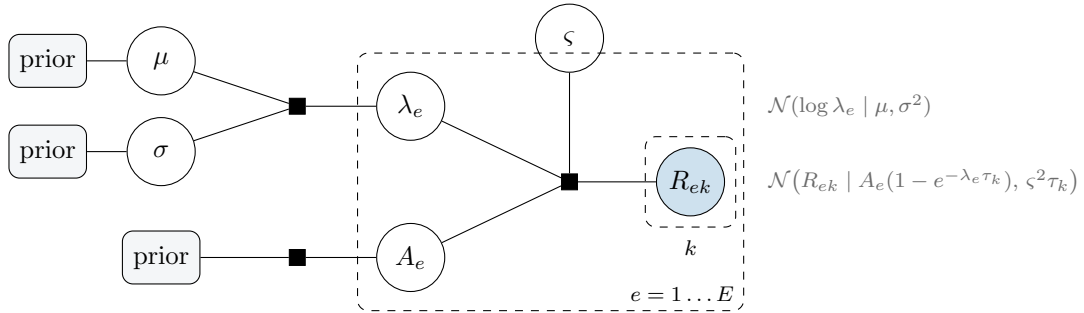


Figure 3: Factor graph. The population factor couples every event’s rate to the shared (μ, σ) , which is the mechanism by which one event’s data informs another’s estimate. The shaded node is observed. The inner plate is the horizon grid: each event contributes several returns, and they are nested, which is why the likelihood must carry their covariance rather than treat them as independent.

6.1 Why this is fitted by sampling, not by message passing

The structure resembles TrueSkill: a latent per-entity quantity, a population prior over it, and shrinkage carrying entities with little data. The resemblance stops at inference.

TrueSkill admits expectation propagation because its factors are chosen so that every message has closed-form moments: a Gaussian prior, a Gaussian performance model, and an ordinal comparison factor whose truncated moments are analytic. The likelihood factor in (3) is nonlinear in λ_e , so its messages have no closed form and a Gaussian approximation to them would be poor near the peak.

More generally, no closed-form posterior exists here. The conditional structure is instructive:

Conditional	Form	Reason
$A_e \mid \lambda_e$	closed form	$1 - e^{-\lambda_e \tau_k}$ is a constant; linear in A_e
$\lambda_e \mid A_e$	none	λ_e enters inside the exponential
$\mu, \sigma \mid \{\lambda_e\}$	closed form given the rates	standard normal hierarchy
Joint	none	the rates are themselves intractable

The joint posterior is a distribution over more than four hundred parameters with no analytic expression. It is explored by Hamiltonian Monte Carlo, which draws from it without requiring it to be written down. Reported quantities are summaries of those draws: a 95% credible interval is the 2.5th and 97.5th percentiles of the sampled values.

7 Priors

Priors are fixed in `PREREGISTRATION.md` before any fit and are not revised afterwards.

The prior on μ is centred on a 10-second half-life, $\log \lambda = -2.7$, with scale 1.5, spanning roughly 0.5 s to 200 s. This is consistent with the sub-minute reactions visible in Figure 1 and wide enough to admit slower incorporation. That the kill-check informs the prior is a soft use of nine events to set a wide belief over two hundred; it is disclosed rather than concealed, and the posterior is dominated by the data at this sample size.

The prior on σ is half-normal, which admits $\sigma \rightarrow 0$. A prior excluding small σ would manufacture heterogeneity between events whether or not any exists.

Prior predictive checks precede any fit: curves simulated from the prior alone must resemble plausible price paths.

8 What is reported

- $\phi(\tau)$ at each horizon with a 95% band. Axes labelled, seconds on a log scale, fraction of eventual move dimensionless.
- The population half-life $\ln 2 / \exp(\mu)$ in seconds, with an interval.
- σ itself. A large σ is a result, not a nuisance: it means incorporation speed is a distribution rather than a constant, which is a more informative finding than its average.
- Release-type and year contrasts on μ , as group effects within the same model.

9 How this could be wrong

The exponential form. Equation (2) assumes a monotone approach. Two panels of Figure 1 appear to overshoot and revert inside the window. A single rate cannot represent that and would report a compromise describing neither phase. It is diagnosed from residual structure across horizons, and a systematic pattern is reported as a finding about the shape of incorporation rather than repaired by refitting.

H is a choice. One hour is arbitrary within an order of magnitude. Sensitivity to $H \in \{30 \text{ min}, 4 \text{ h}\}$ is a preregistered check.

The eventual move is not the release. $R_e(H)$ contains all news in the window.

Overlapping horizons. $R_e(\tau_k)$ is nested in $R_e(\tau_{k+1})$, so the observations are mechanically dependent. Treating them as independent narrows the intervals, and that error presents as a sharper result.

Contamination. H must contain no second scheduled release. FOMC statements fail this: the chair’s press conference begins at 14:30 ET, inside a one-hour window, at every meeting since 2019. FOMC is therefore excluded from the primary sample and retained as a robustness check (ADR 0002).

The full list, in a reviewer’s phrasing and including the objections with no answer, is `docs/limitations.md`.

10 Validation before any real data

The model is fitted to simulated data with a known λ and must recover it before release data is touched. A model that cannot recover a rate it generated itself will not recover one from Binance.

This is simulation-based calibration. It distinguishes an unidentified parameter from a result, which is a distinction that is expensive to discover late.

No estimation code is written before `PREREGISTRATION.md` is merged. The public commit timestamp establishing that the analysis plan predates the results is the principal artifact of this repository.